

Notes on Gauge Theory

Bob

Contents

0	Prologue: Electromagnetism and $U(1)$ Gauge Theory	2
0.1	Maxwell's equations	2
0.2	$U(1)$ gauge theory	5
1	Affine Connections	7
1.1	Levi-Civita Connections	7
1.1.1	Covariant Derivative	7
1.1.2	Connection 1-form	8
1.2	General Affine Connections	11
1.2.1	Covariant Derivative	11
1.2.2	Connection 1-form	13
1.2.3	Horizontal Distribution	15
	Exercises	20

0 Prologue: Electromagnetism and $U(1)$ Gauge Theory

0.1 Maxwell's equations

We begin the story with Maxwell's equations. Recall that Maxwell's equations in vacuum are

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases}$$

Here

$$\begin{aligned} \mathbf{E} &= (E_x, E_y, E_z) \quad \text{is the electric field,} \\ \mathbf{B} &= (B_x, B_y, B_z) \quad \text{is the magnetic field.} \end{aligned}$$

Each component is a function of the spacetime coordinates (t, x, y, z) .

Below we will list some concepts and notations commonly used in physics:

- 4-vector potential A_μ ¹

$$A_\mu = (\phi, A_x, A_y, A_z),$$

where ϕ is the *scalar potential* and $\mathbf{A} = (A_x, A_y, A_z)$ is the *vector potential*.

- Electromagnetic field strength tensor $F_{\mu\nu}$

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

- Relation between A_μ and $F_{\mu\nu}$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

- Gauge transformation of the 4-vector potential

$$A_\mu \mapsto A_\mu + \partial_\mu \chi \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \operatorname{grad} \chi. \end{cases}$$

where χ is an arbitrary smooth function. One can check that this transformation leaves the electromagnetic field strength tensor $F_{\mu\nu}$ invariant.

¹ $\mu = 0, 1, 2, 3$ are the spacetime indices, where 0 corresponds to the time coordinate t , and 1, 2, 3 correspond to the spatial coordinates x, y, z .

²Here we use the notation $\partial_0 := \frac{\partial}{\partial t}$, $\partial_1 := \frac{\partial}{\partial x}$, $\partial_2 := \frac{\partial}{\partial y}$, $\partial_3 := \frac{\partial}{\partial z}$.

Although Maxwell's equations are already quite concise, we can still transform them into an even more elegant expression using the language of differential forms.

Let $\mathbb{R}^{3,1}$ be the spacetime with coordinates (t, x, y, z) , equipped with the Minkowski metric

$$ds^2 = dt^2 - dx^2 - dy^2 - dz^2.$$

For simplicity, we set the speed of light $c = 1$.

The Hodge star operator $*$ on $\Omega^*(\mathbb{R}^{3,1})$ is defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle dt \wedge dx \wedge dy \wedge dz, \quad \forall \alpha, \beta \in \Omega^*(\mathbb{R}^{3,1}).$$

Explicitly, we have

$$\begin{aligned} *1 &= dt \wedge dx \wedge dy \wedge dz, \\ *dt &= dx \wedge dy \wedge dz, \\ *dx &= dt \wedge dy \wedge dz, \\ *dy &= dt \wedge dz \wedge dx, \\ *dz &= dt \wedge dx \wedge dy, \\ *(dt \wedge dx) &= -dy \wedge dz, \\ *(dt \wedge dy) &= -dz \wedge dx, \\ *(dt \wedge dz) &= -dx \wedge dy, \\ *(dx \wedge dy) &= dt \wedge dz, \\ *(dy \wedge dz) &= dt \wedge dx, \\ *(dz \wedge dx) &= dt \wedge dy, \\ *(dt \wedge dx \wedge dy) &= dz, \\ *(dt \wedge dy \wedge dz) &= dx, \\ *(dt \wedge dz \wedge dx) &= dy, \\ *(dx \wedge dy \wedge dz) &= dt, \\ *(dt \wedge dx \wedge dy \wedge dz) &= -1. \end{aligned}$$

It is direct to check that

$$*^2 = (-1)^{p+1} \quad \text{on } \Omega^p(\mathbb{R}^{3,1}).$$

Now we can introduce the electromagnetic field strength 2-form

$$F = -dt \wedge (E_x dx + E_y dy + E_z dz) + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$$

Then

$$\begin{aligned} dF &= dt \wedge \left[(\partial_x E_y - \partial_y E_x) dx \wedge dy + (\partial_y E_z - \partial_z E_y) dy \wedge dz + (\partial_z E_x - \partial_x E_z) dz \wedge dx \right] \\ &\quad + dt \wedge \left(\partial_t B_x dy \wedge dz + \partial_t B_y dz \wedge dx + \partial_t B_z dx \wedge dy \right) \\ &\quad + \left(\partial_x B_x + \partial_y B_y + \partial_z B_z \right) dx \wedge dy \wedge dz, \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \end{cases} \iff dF = 0.$$

On the other hand,

$$*F = E_x dy \wedge dz + E_y dz \wedge dx + E_z dx \wedge dy + dt \wedge (B_x dx + B_y dy + B_z dz).$$

So

$$\begin{aligned} d * F &= -dt \wedge \left[(\partial_x B_y - \partial_y B_x) dx \wedge dy + (\partial_z B_x - \partial_x B_z) dz \wedge dx + (\partial_y B_z - \partial_z B_y) dy \wedge dz \right] \\ &\quad + dt \wedge \left(\partial_t E_x dy \wedge dz + \partial_t E_y dz \wedge dx + \partial_t E_z dx \wedge dy \right) \\ &\quad + \left(\partial_x E_x + \partial_y E_y + \partial_z E_z \right) dx \wedge dy \wedge dz. \end{aligned}$$

Thus

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff d * F = 0.$$

Since $*$: $\Omega^p(\mathbb{R}^{3,1}) \rightarrow \Omega^{4-p}(\mathbb{R}^{3,1})$ is an isomorphism,

$$d * F = 0 \iff *d * F = 0.$$

Therefore,

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \begin{cases} dF = 0, \\ d * F = 0. \end{cases}$$

Here $d^* = *d*$ is the dual of d .

In Riemannian geometry, a p -form α is called *harmonic* if $d\alpha = 0$ and $d^*\alpha = 0$. Thus Maxwell's equations in vacuum corresponds to a harmonic 2-form F .

Poincaré lemma says all closed forms on $\mathbb{R}^{3,1}$ are exact. So if $dF = 0$, there exists a 1-form A such that

$$F = dA.$$

If we write

$$A = \phi dt + A_x dx + A_y dy + A_z dz,$$

Then

$$F = dA \implies \begin{cases} E_x = \partial_x \phi - \partial_t A_x, \\ E_y = \partial_y \phi - \partial_t A_y, \\ E_z = \partial_z \phi - \partial_t A_z, \\ B_x = \partial_y A_z - \partial_z A_y, \\ B_y = \partial_z A_x - \partial_x A_z, \\ B_z = \partial_x A_y - \partial_y A_x. \end{cases} \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

It is easy to see that A is not unique. In fact, if we transform $A \mapsto A + d\chi$ for some smooth function χ , then

$$F \mapsto dA + d^2\chi = dA = F.$$

Explicitly,

$$A \mapsto A + d\chi \implies \begin{cases} \phi \mapsto \phi + \partial_t \chi, \\ A_x \mapsto A_x + \partial_x \chi, \\ A_y \mapsto A_y + \partial_y \chi, \\ A_z \mapsto A_z + \partial_z \chi. \end{cases} \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \nabla \chi. \end{cases}$$

We can summarize the above discussion in a table:

Physics	Differential Geometry
4-vector potential A_μ	1-form A
Electromagnetic field strength tensor $F_{\mu\nu}$	2-form F
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F = dA$
Gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$	$A \mapsto A + d\chi$
Maxwell's equations	$dF = 0, d^*F = 0$

0.2 $U(1)$ gauge theory

Let $\psi(x)$ be a complex-valued function on $\mathbb{R}^{3,1}$, which represents the wave function of a matter field. Here x denotes the spacetime coordinates. In physics, $|\psi(x)|^2$ represents the probability amplitude of finding the particle at a given point in spacetime. If we transform

$$\psi(x) \mapsto \psi'(x) = e^{i\chi} \psi(x)$$

for some real constant χ . This transformation is called a *global $U(1)$ gauge transformation*. Since the probability amplitude $|\psi'(x)|^2 = |\psi(x)|^2$ remains the same, the physics is invariant under global $U(1)$ gauge transformations.

Now we want to promote the global $U(1)$ gauge symmetry to a local one. That is, we want the physics to be invariant under the transformation

$$\psi(x) \mapsto \psi'(x) = e^{i\chi(x)} \psi(x)$$

for some smooth function $\chi \in C^\infty(\mathbb{R}^{3,1})$.

For some physical reason, we want the derivative of ψ to be covariant under transformations. This holds in the case of global $U(1)$ gauge transformations, since

$$d\psi \mapsto d\psi' = e^{i\chi} d\psi.$$

However, in the case of local $U(1)$ gauge transformations,

$$d\psi \mapsto d\psi' = e^{i\chi(x)} d\psi + (de^{i\chi(x)})\psi.$$

The extra term $(de^{i\chi(x)})\psi$ breaks the covariance of the derivative.

To fix this problem, we introduce a 1-form A and define the *covariant derivative* by

$$\nabla\psi = d\psi + iA\psi.$$

Under the local $U(1)$ gauge transformation,

$$\nabla \mapsto \nabla' = d + iA',$$

we want $\nabla\psi$ to be covariant, i.e.

$$\nabla\psi \mapsto \nabla'\psi' = e^{i\chi(x)}\nabla\psi.$$

Then by

$$\begin{aligned}\nabla'\psi' &= d\psi' + iA'\psi' = e^{i\chi(x)}d\psi + ie^{i\chi(x)}(A' + d\chi)\psi, \\ e^{i\chi(x)}\nabla\psi &= e^{i\chi(x)}(d\psi + iA\psi) = e^{i\chi(x)}d\psi + ie^{i\chi(x)}A\psi,\end{aligned}$$

we must have

$$A' = A - d\chi.$$

Ignoring the sign, this is exactly the gauge transformation of the electromagnetic potential A in electromagnetism. Thus, the electromagnetic potential A can be interpreted as a gauge field that ensures the covariance of the derivative under local $U(1)$ gauge transformations.

1 Affine Connections

1.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

1.1.1 Covariant Derivative

Definition 1.1. A **connection** ∇ on a manifold M is a map

$$\begin{aligned}\nabla : C^\infty(M, TM) \times C^\infty(M, TM) &\rightarrow C^\infty(M, TM) \\ (Y, X) &\mapsto \nabla_Y X\end{aligned}$$

satisfying the following properties:

$$\begin{aligned}\nabla_{f_1 Y_1 + f_2 Y_2} X &= f_1 \nabla_{Y_1} X + f_2 \nabla_{Y_2} X, \\ \nabla_Y (\lambda_1 X_1 + \lambda_2 X_2) &= \lambda_1 \nabla_Y X_1 + \lambda_2 \nabla_Y X_2, \\ \nabla_Y f X &= df(Y) \cdot X + f \nabla_Y X.\end{aligned}$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 1.2. The **torsion** of a connection ∇ is defined as the tensor

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y].$$

The **curvature** of a connection ∇ is defined as the tensor

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Remark. The torsion T can only be defined for connections on the tangent bundle, since it involves both $\nabla_X Y$ and $\nabla_Y X$.

Definition 1.3. A connection ∇ of a Riemannian manifold (M, g) is called a **Levi-Civita connection** if it is torsion-free and compatible with the metric g , i.e.

$$\begin{aligned}\nabla_X Y - \nabla_Y X &= [X, Y], \\ X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z).\end{aligned}$$

Exterior covariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned}\nabla : C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M \otimes TM) \\ X &\mapsto \nabla X\end{aligned}$$

where ∇X is a 1-form taken value in TM^1 and $(\nabla X)(Y) := \nabla_Y X$. ∇ is not a $C^\infty(M)$ -linear map, instead,

$$\nabla(fX) = df \otimes X + f \nabla X.$$

¹One can simply view ∇X as a $(1, 1)$ -tensor field.

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, TM) &\rightarrow \Omega^{*+1}(M, TM)^1 \\ \eta &\mapsto \nabla\eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla\eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, TM),$$

For example, let $\eta = \alpha \otimes X$ be a section of $\Omega^r(M, TM)$, then

$$\nabla\eta = \nabla(\alpha \otimes X) = d\alpha \otimes X + (-1)^r \alpha \wedge \nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla \circ \nabla, \nabla^3, \dots, \nabla^k$, and we can prove the following property:

Proposition 1.4. $\nabla^2 : \Omega^*(M, TM) \rightarrow \Omega^{*+2}(M, TM)$ is an $\Omega^*(M)$ -linear map, i.e. for any $\alpha \in \Omega^*(M), \eta \in \Omega^*(M, TM)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2\eta.$$

Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \text{End}(TM))$, which is called the **curvature** of the connection ∇ .

Proof.

$$\begin{aligned}\nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge d\nabla\eta) \\ &= d^2\alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge d\nabla\eta + (-1)^q d\alpha \wedge \nabla\eta + \alpha \wedge \nabla^2\eta \\ &= \alpha \wedge \nabla^2\eta.\end{aligned}$$

□

Unlike the usual exterior derivative d , we have $\nabla^2 \neq 0$ in general. The following proposition explains why ∇^2 is called the curvature of ∇ :

Proposition 1.5. For $X, Y, Z \in C^\infty(M, TM)$, we have

$$(\nabla^2 Z)(X, Y) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Proof. We will prove it after introducing the connection 1-forms. □

1.1.2 Connection 1-form

Locally, we always have a frame. Let $\{e_i\}$ be a local frame of TM on an open set $U \subset M$, and $\{\theta^i\}$ be the dual frame. then we can write

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k,$$

where Γ_{ij}^k are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega_j^k := \Gamma_{ij}^k \theta^i,$$

¹ $\Omega^r(M, TM) := C^\infty(M, \wedge^r T^*M \otimes TM).$

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)^1$, then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_n) = (e_1, \dots, e_n) \omega = (e_1, \dots, e_n) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \cdots & \omega_n^n \end{pmatrix}^2$$

For any vector field $X = X^i e_i$, we have

$$\begin{aligned} \nabla X &= \nabla \left((e_1, \dots, e_n) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right) \\ &= \left(\nabla(e_1, \dots, e_n) \right) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + (e_1, \dots, e_n) \begin{pmatrix} dX^1 \\ \vdots \\ dX^n \end{pmatrix} \\ &= (e_1, \dots, e_n) \left(d \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + \omega \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right). \end{aligned}$$

Thus under a local frame, ∇ can be expressed as

$$\nabla = d + \omega.$$

We can compute the curvature in terms of the connection 1-form ω :

$$\begin{aligned} \nabla^2(e_1, \dots, e_n) &= \nabla(\nabla(e_1, \dots, e_n)) = \nabla((e_1, \dots, e_n) \omega) \\ &= (\nabla(e_1, \dots, e_n)) \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) \omega \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n) (d\omega + \omega \wedge \omega). \end{aligned}$$

Define

$$\Omega := d\omega + \omega \wedge \omega, \tag{1.1}$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 1.6. *Let $R(e_i, e_j)e_k = R_{ijk}^l e_l$, then*

$$\Omega_k^l(e_i, e_j) = R_{ijk}^l.$$

¹The upper index k is the row index, and the lower index j is the column index.

²This is an awkward notation. Since we arrange the frame as row vectors so matrix ω should multiply on the right. For a vector field X and a differential forms $\alpha \in \Omega^q(M)$, we have $X \otimes \alpha = \alpha \otimes X$. For a general $\eta \in \Omega^p(M, TM)$, we have $\eta \otimes \alpha = (-1)^{pq} \alpha \otimes \eta$.

Proof. Since $\Omega_k^l = d\omega_k^l + \omega_m^l \wedge \omega_k^m$,

$$\begin{aligned} (d\omega_k^l)(e_i, e_j)e_l &= e_i(\omega_k^l(e_j))e_l - e_j(\omega_k^l(e_i))e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}(\omega_k^l(e_j)e_l) - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega_k^l(e_i)e_l) + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}\nabla_{e_j}e_k - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l, \end{aligned}$$

and

$$\begin{aligned} (\omega_m^l \wedge \omega_k^m)(e_i, e_j)e_l &= \omega_m^l(e_i)\omega_k^m(e_j)e_l - \omega_m^l(e_j)\omega_k^m(e_i)e_l \\ &= \omega_k^m(e_j)\nabla_{e_i}e_m - \omega_k^m(e_i)\nabla_{e_j}e_m. \end{aligned}$$

Thus

$$\Omega_k^l(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i, e_j]}e_k = R(e_i, e_j)e_k. \quad \square$$

Corollary 1.7. *This proposition immediately implies Prop 1.5 since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega_k^l(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

Let $\{\tilde{e}_j\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_j\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n) a,$$

then

$$\begin{aligned} (\tilde{e}_1, \dots, \tilde{e}_n)\tilde{\omega} &= \nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla((e_1, \dots, e_n) a) \\ &= (\nabla(e_1, \dots, e_n))a + (e_1, \dots, e_n)da \\ &= (e_1, \dots, e_n)\omega a + (e_1, \dots, e_n)da \\ &= (\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + da). \end{aligned}$$

Thus

$$\tilde{\omega} = a^{-1}da + a^{-1}\omega a. \quad (1.2)$$

Applying the transformation formula (1.2) to (1.1), we get

$$\begin{aligned} \tilde{\Omega} &= d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega} \\ &= d(a^{-1}da + a^{-1}\omega a) + (a^{-1}da + a^{-1}\omega a) \wedge (a^{-1}da + a^{-1}\omega a) \\ &= da^{-1} \wedge da + da^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a, \end{aligned}$$

since

$$a^{-1}a = I \implies da^{-1}a + a^{-1}da = 0 \implies da^{-1} = -a^{-1}da a^{-1},$$

we have

$$\begin{aligned} \tilde{\Omega} &= -a^{-1}da a^{-1} \wedge da - a^{-1}da a^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a \\ &= a^{-1}(d\omega + \omega \wedge \omega)a = a^{-1}\Omega a. \end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \quad (1.3)$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

1.2 General Affine Connections

Let $\pi : E \rightarrow M$ be a rank m vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

1.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 1.8. An (affine) connection ∇ on a vector bundle E is a map

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, E) &\rightarrow C^\infty(M, E) \\ (X, s) &\mapsto \nabla_X s \end{aligned}$$

satisfying

$$\begin{aligned} \nabla_{f_1 X_1 + f_2 X_2} s &= f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s, \\ \nabla_X (\lambda_1 s_1 + \lambda_2 s_2) &= \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2, \\ \nabla_X (fs) &= df(X) \cdot s + f \nabla_X s, \end{aligned}$$

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned} \nabla : \Omega^*(M, E) &\rightarrow \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla \eta, \end{aligned}$$

satisfying the Leibniz rule: for any $\alpha \in \Omega^q(M)$, $\eta \in \Omega^*(M, E)$,

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta.$$

Remark. The reason why we call ∇ an **affine** connection is that given two connections ∇_1 and ∇_2 on E , $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Remark. The existence of connections on E can be proved by using partitions of unity, which we left as an exercise.

Proposition 1.9. Let ∇_1 and ∇_2 be two connections on E , then $\nabla_1 - \nabla_2$ is $C^\infty(M)$ -linear in s , thus defines a section $A \in \Omega^1(M, \text{End}(E))$. Conversely, for any $A \in \Omega^1(M, \text{End}(E))$ and a connection ∇ on E , $\nabla + A$ is also a connection on E . *Consequently, the space of connections on E is an affine space modeled on $\Omega^1(M, \text{End}(E))$.*

Proof. For $s \in C^\infty(M, E)$, $f \in C^\infty(M)$,

$$(\nabla_1 - \nabla_2)(fs) = df \otimes s + f \nabla_1 s - df \otimes s - f \nabla_2 s = f(\nabla_1 - \nabla_2)(s),$$

thus $\nabla_1 - \nabla_2$ is $C^\infty(M)$ -linear in s . Conversely, for $A \in \Omega^1(M, \text{End}(E))$ and a connection ∇ on E ,

$$(\nabla + A)(fs) = \nabla(fs) + A(fs) = df \otimes s + f \nabla s + f As = df \otimes s + f(\nabla + A)(s). \quad \square$$

Curvature of connections

Proposition 1.10. $\nabla^2 : \Omega^*(M, E) \rightarrow \Omega^{*+2}(M, E)$ is an $\Omega^*(M)$ -linear map, i.e., for any $\alpha \in \Omega^*(M), \eta \in \Omega^*(M, E)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta.$$

Proof. For $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\begin{aligned} \nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta) \\ &= d^2 \alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge \nabla \eta + (-1)^q d\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ &= \alpha \wedge \nabla^2 \eta. \end{aligned}$$

□

Definition 1.11. The curvature of ∇ is defined as $R := \nabla^2 \in \Omega^2(M, \text{End}(E))$.

Metrics and metric compatibility

Definition 1.12. Let E be a (complex) vector bundle over M . A **(Hermitian) metric** on E is a smooth section g of $E^* \otimes E^*$ such that for any $x \in M$, g_x is a (Hermitian) inner product $\langle \cdot, \cdot \rangle_{E_x}$ on the fibre E_x .

Definition 1.13. A connection ∇ on E is called **compatible with the metric g** if for any $X \in C^\infty(M, TM)$ and $s_1, s_2 \in C^\infty(M, E)$,

$$X\langle s_1, s_2 \rangle = \langle \nabla_X s_1, s_2 \rangle + \langle s_1, \nabla_X s_2 \rangle.$$

Remark. For a (complex) vector bundle E over M , we can always construct a (Hermitian) metric on E by using partitions of unity.

Remark. Given a (Hermitian) metric g on E , we can always construct a connection ∇ compatible with g by using partitions of unity. Note that the connection compatible with g is not unique.

Proposition 1.14. Let ∇_1 and ∇_2 be two connections on E compatible with metric g , then $\nabla_1 - \nabla_2$ is skew-symmetric w.r.t. g , i.e., $\nabla_1 - \nabla_2 \in \Omega^1(M, \mathfrak{so}(E, g))$. Conversely, for any $A \in \Omega^1(M, \mathfrak{so}(E, g))$ and a connection ∇ compatible with g , $\nabla + A$ is also compatible with g . *Consequently, the space of connections on E compatible with g is an affine space modeled on $\Omega^1(M, \mathfrak{so}(E, g))$.*

Proof. For $s_1, s_2 \in C^\infty(M, E), X \in C^\infty(M, TM)$,

$$\begin{aligned} &\begin{cases} X\langle s_1, s_2 \rangle = \langle \nabla_{1,X} s_1, s_2 \rangle + \langle s_1, \nabla_{1,X} s_2 \rangle \\ X\langle s_1, s_2 \rangle = \langle \nabla_{2,X} s_1, s_2 \rangle + \langle s_1, \nabla_{2,X} s_2 \rangle \end{cases} \\ &\implies \langle (\nabla_1 - \nabla_2)_X s_1, s_2 \rangle + \langle s_1, (\nabla_1 - \nabla_2)_X s_2 \rangle = 0. \end{aligned}$$

Conversely, for $A \in \Omega^1(M, \mathfrak{so}(E, g))$ and a connection ∇ compatible with g ,

$$\begin{aligned} &\langle (\nabla + A)_X s_1, s_2 \rangle + \langle s_1, (\nabla + A)_X s_2 \rangle \\ &= \langle \nabla_X s_1, s_2 \rangle + \langle s_1, \nabla_X s_2 \rangle + \langle A(X)s_1, s_2 \rangle + \langle s_1, A(X)s_2 \rangle \\ &= X\langle s_1, s_2 \rangle. \end{aligned}$$

□

1.2.2 Connection 1-form

Let E be a rank m vector bundle over M , and $\{e_i\}$ be a local frame of E on an open set $U \subset M$. Define ω_j^k by

$$\nabla_X e_j = \omega_j^k(X) e_k,$$

this 1-form matrix $\omega = (\omega_j^k)$ is called **the connection 1-form** of ∇ w.r.t. $\{e_i\}$. Then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_m) = (e_1, \dots, e_m) \omega = (e_1, \dots, e_m) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^m \\ \vdots & & \vdots \\ \omega_m^1 & \cdots & \omega_m^m \end{pmatrix}.$$

Thus for any section $s = s^i e_i$, we have

$$\begin{aligned} \nabla s &= \nabla \left((e_1, \dots, e_m) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right) = \left(\nabla(e_1, \dots, e_m) \right) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + (e_1, \dots, e_m) \begin{pmatrix} ds^1 \\ \vdots \\ ds^m \end{pmatrix} \\ &= (e_1, \dots, e_m) \left(d \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + \omega \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right). \end{aligned}$$

So under a local frame, ∇ can be expressed as

$$\nabla = d + \omega.$$

Proposition 1.15. Suppose $\{\tilde{e}_i\}$ are another local frame of E on U , and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_i\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_m) = (e_1, \dots, e_m) a,$$

Let $\tilde{\omega}$ be the connection 1-form w.r.t. $\{\tilde{e}_i\}$, then the connection 1-forms transform as

$$\tilde{\omega} = a^{-1} da + a^{-1} \omega a.$$

A local frame of E is the same as a local trivialization. Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of E , $g_{\alpha\beta}$ be the transition functions. A section $s \in C^\infty(M, E)$ possesses a local representative s_α , which is an m -tuple of smooth functions on U_α . Also, ∇s possesses a local representative $(\nabla s)_\alpha$, which is an m -tuple of 1-forms on U_α . Just as in the preceding paragraph, we can write

$$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha, \tag{1.4}$$

where ω_α is an $m \times m$ matrix of 1-forms on U_α .

Since the connection is globally defined, the local representatives of ∇s must satisfy

$$ds_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1} (ds_\alpha + \omega_\alpha s_\alpha) \quad \text{on } U_\alpha \cap U_\beta.$$

Apply $s_\alpha = g_{\alpha\beta} s_\beta$ to the above equation,

$$ds_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1} (d(g_{\alpha\beta} s_\beta) + \omega_\alpha g_{\alpha\beta} s_\beta)$$

$$\begin{aligned}
&= g_{\alpha\beta}^{-1}((dg_{\alpha\beta})s_\beta + g_{\alpha\beta}ds_\beta + \omega_\alpha g_{\alpha\beta}s_\beta) \\
&= ds_\beta + \left(g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}\right)s_\beta,
\end{aligned}$$

Thus

$$\omega_\beta = g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}. \quad (1.5)$$

The above discussion can be summarized as the following theorem:

Theorem 1.16. *Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of E , with transition functions $g_{\alpha\beta}$, and ω_α be an $m \times m$ matrix of 1-forms on U_α satisfying the transformation formula (1.5). This information uniquely determines a connection ∇ on E .*

Let (E, g) be a real vector bundle equipped with a metric, we can choose the local frame $\{e_i\}$ on U to be orthonormal, then for any $X \in C^\infty(U, TU)$,

$$0 = X\langle e_i, e_j \rangle = \langle \omega_i^k(X)e_k, e_j \rangle + \langle e_i, \omega_j^k(X)e_k \rangle = \omega_j^i(X) + \omega_i^j(X),$$

so the 1-form matrix ω is skew-symmetric, i.e., $\omega \in \Omega^1(U, \mathfrak{so}(m))$.

Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of (E, g) . We can assume

$$\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^m$$

is isometric on each fibre, where the metric on $U_\alpha \times \mathbb{R}^m$ is Euclidean metric on each fibre. Then the transition functions $g_{\alpha\beta}$ take values in $O(m)$, and each ω_α is in $\Omega^1(U_\alpha, \mathfrak{so}(m))$.

If E is a complex vector bundle and g is a Hermitian metric. Then all the discussion above still holds, except that ω_α is in $\Omega^1(U_\alpha, \mathfrak{u}(m))$ and $g_{\alpha\beta}$ takes values in $U(m)$.

Example 1.17. Let $\pi : L \rightarrow M$ be a complex line bundle, g be a Hermitian metric on L and ∇ be a connection on L compatible with g . Choose a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of L such that $\psi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{C}$ is an isometry. Then $g_{\alpha\beta}$ takes values in $U(1)$, so we can write $g_{\alpha\beta} = e^{i\theta_{\alpha\beta}}$ for some real-valued function $\theta_{\alpha\beta}$ on $U_\alpha \cap U_\beta$. Meanwhile, ω_α takes values in $\mathfrak{u}(1)$, so we can write $\omega_\alpha = iA_\alpha$ for some real-valued 1-form A_α on U_α . For any section s of L , on U_α , we have

$$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha, \quad \omega_\alpha = iA_\alpha,$$

On $U_\alpha \cap U_\beta$,

$$\omega_\beta = g_{\alpha\beta}^{-1}dg_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta} = e^{-i\theta_{\alpha\beta}}de^{i\theta_{\alpha\beta}} + e^{-i\theta_{\alpha\beta}}\omega_\alpha e^{i\theta_{\alpha\beta}} = \omega_\alpha + id\theta_{\alpha\beta} \left(= \omega_\alpha - id\theta_{\beta\alpha} \right)$$

Recall $U(1)$ -gauge theory in physics, the following table shows the correspondence between concepts in physics and differential geometry:

Physics	Differential Geometry
wave function ψ	section s of complex line bundle L
$\psi \mapsto \psi' = e^{i\chi}\psi$	$s_\beta = g_{\beta\alpha}s_\alpha$
covariant derivative	connection ∇ on L
$D'\psi' = e^{i\chi}D\psi$	$(\nabla s)_\beta = g_{\beta\alpha}(\nabla s)_\alpha$ ¹
$D = d + iA$	$(\nabla s)_\alpha = ds_\alpha + \omega_\alpha s_\alpha$
$A' = A - d\chi$	$\omega_\beta = \omega_\alpha - id\theta_{\beta\alpha}$

¹That is to say, ∇s is globally defined.

1.2.3 Horizontal Distribution

In this subsection, we give another interpretation of a connection on a vector bundle, namely a horizontal distribution, and prove that it is equivalent to the definition given in terms of covariant derivatives.

Let $\pi : E \rightarrow M$ be a rank m vector bundle, and $p \in E_x$ be a point in the fibre over $x \in M$. The projection π induces a linear map $\pi_{*p} : T_p E \rightarrow T_x M$. We define $\mathcal{V}_p := \ker \pi_{*p}$, which is called the **vertical tangent space** at p . Thus we have a short exact sequence

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_p E \xrightarrow{\pi_*} T_{\pi(p)} M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_p E - \dim T_{\pi(p)} M = m.$$

The kernel of π_* form a subbundle of TE . We call it the **vertical subbundle** of TE and denote it by $\mathcal{V}$.

Proposition 1.18. *For each $p \in E$, there is an isomorphism $E_{\pi(p)} \cong \mathcal{V}_p$, which induces an isomorphism $\pi^* E \cong \mathcal{V}$.*

Proof. A point $(p, p') \in \pi^* E$ must satisfy $\pi(p) = \pi(p')$, so we can define a curve $\gamma(t) = p + tp'$ in E . Then $\gamma'(0) \in T_p E$ gives a tangent vector at p , this gives an injective morphism

$$\pi^* E \hookrightarrow TE, \quad (p, p') \mapsto (p, \gamma'(0)).$$

So we can view $\pi^* E$ as a subbundle of TE . Moreover, since $\gamma(t) \in E_{\pi(p)}$, we have $\pi(\gamma(t)) = \pi(p)$, thus $\pi_{*p}(\gamma'(0)) = 0$. Therefore, $\pi^* E \subset \ker \pi_*$.

Since π_{*p} is a surjective linear map, we have

$$\dim \ker \pi_{*p} = \dim T_p E - \dim T_{\pi(p)} M = m = (\pi^* E)_p.$$

Thus $\pi^* E = \ker \pi_*$. □

For any $p \in E$, we can define a map

$$\iota_p : E_{\pi(p)} \rightarrow \mathcal{V}_p \subset T_p E, \quad p' \mapsto \gamma'(0),$$

where $\gamma(t) = p + tp'$ is a curve in E . Then ι_p is an isomorphism between $E_{\pi(p)}$ and $\mathcal{V}_p$.

Let $\pi^* TM$ be the pull-back of TM by π , then $\pi_* : TE \rightarrow TM$ can be viewed as a morphism $\tilde{\pi}_* : TE \rightarrow \pi^* TM$. The above proposition implies the following short exact sequence of vector bundles over E :

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \xrightarrow{\tilde{\pi}_*} \pi^* TM \longrightarrow 0$$

In smooth category, all short exact sequences of vector bundles split, so we can find a subbundle $\mathcal{H} \hookrightarrow TE$ such that

$$TE = \mathcal{H} \oplus \mathcal{V}, \quad \tilde{\pi}_* : \mathcal{H} \rightarrow \pi^* TM \text{ is an isomorphism.}$$

On the vector bundle E , we have two natural operations: scalar multiplication and addition.

- The scalar multiplication:

$$m : \mathbb{R} \times E \rightarrow E, \quad m(\lambda, p) = \lambda p.$$

When fixing λ , we get a map $m_\lambda : E \rightarrow E$, $m_\lambda(p) = \lambda p$, which is the scalar multiplication by λ .

- The addition:

$$a : E \times_M E \rightarrow E, \quad a(p, q) = p + q,$$

where $E \times_M E = \{(p, q) \in E \times E : \pi(p) = \pi(q)\}$.

Definition 1.19. A **horizontal distribution** of E is a smooth subbundle $\mathcal{H} \subset TE$ satisfying the following properties

1. $\tilde{\pi}_* : \mathcal{H} \rightarrow \pi^*TM$ is an isomorphism, or equivalently, there exists a splitting map $k : \pi^*TM \rightarrow TE$ such that $\tilde{\pi}_* \circ k = \text{id}$.

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \xrightleftharpoons[k]{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

2. $\mathcal{H}$ is invariant under the scalar multiplication on E , i.e. for any $\lambda \in \mathbb{R}$, $p \in E$,

$$m_{\lambda*} \mathcal{H}_p = \mathcal{H}_{\lambda p}.$$

3. $\mathcal{H}$ is invariant under the addition on E , i.e. for any $(p, q) \in E \times_M E$,

$$a_{*(p,q)}(\mathcal{H}_p, \mathcal{H}_q) = \mathcal{H}_{p+q}.$$

The last two conditions are saying that $\mathcal{H}$ is compatible with the linear structure of E .

There is no canonical way to choose a horizontal distribution, so a horizontal distribution is an extra structure on E .

In what follows, we reveal that covariant derivatives and horizontal distributions can actually determine each other.

Covariant Derivative determines Horizontal Distribution

Let ∇ be a connection on E . Given any smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = x$, we say $\tilde{\gamma} : (-\varepsilon, \varepsilon) \rightarrow E$ is a **lift** of γ if $\pi(\tilde{\gamma}(t)) = \gamma(t)$ for all t .

Definition 1.20. A lift $\tilde{\gamma}$ of γ is called a **horizontal lift** if

$$\nabla_{\gamma'(t)} \tilde{\gamma}(t) = 0, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Example 1.21. In the case of a Riemannian manifold (M, g) , the horizontal lift is exactly the parallel transport.

Proposition 1.22. For any (piecewise) smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ and any point $p \in E_{\gamma(0)}$, there exists a unique horizontal lift $\tilde{\gamma}$ of γ starting at p .

Sketch of proof. The proof is exactly the same as the existence and uniqueness of geodesics. The outline of the proof is to write the equations as a system of ODEs in a local frame; the conclusion then follows from the existence and uniqueness theory of linear ODEs. $\square$

For any vector $u \in T_x M$ and any point $p \in E_x$, we can find a curve γ such that $\gamma(0) = x$ and $\gamma'(0) = u$, the horizontal lift $\tilde{\gamma}$ gives a tangent vector $\tilde{\gamma}'(0) \in T_p E$.

Fact 1.1. $\tilde{\gamma}'(0) \in T_p E$ is independent of the choice of γ .

Sketch of proof. By solving the ODEs, we know that $\tilde{\gamma}'(0)$ depends only on the derivative of γ at 0, which is $\gamma'(0) = u$. $\square$

So there is a well-defined map

$$T_{\pi(p)} M \rightarrow T_p E, \quad (u^h)_p := \tilde{\gamma}'(0).$$

We call $(u^h)_p$ the **horizontal lift** of u at p . By definition,

$$\pi(\tilde{\gamma}(t)) = \gamma(t) \implies \pi_{*p}((u^h)_p) = \pi_{*p}(\tilde{\gamma}'(0)) = \gamma'(0) = u.$$

So we obtain a linear injection from $T_{\pi(p)} M$ to $T_p E$, whose image forms a distribution on E denoted by $\mathcal{H}$, i.e.,

$$\mathcal{H} := \{(p, (u^h)_p) \mid p \in E, u \in T_{\pi(p)} M\}.$$

Fact 1.2. $\mathcal{H}$ is a smooth subbundle of TE .

Sketch of proof. This fact relies on the smooth dependence of solutions of linear ODEs on initial conditions and parameters. $\square$

Fact 1.3. $\mathcal{H}$ is invariant under the scalar multiplication and addition on E .

Proof. Let $p, q \in E$ such that $\pi(p) = \pi(q)$, $\lambda \in \mathbb{R}$, and γ be a curve such that $\gamma(0) = \pi(p)$. Let $\tilde{\gamma}_p, \tilde{\gamma}_q$ be the horizontal lift of γ starting at p, q respectively, then by the uniqueness of horizontal lift, $\lambda \tilde{\gamma}_p$ is the horizontal lift of γ starting at λp , and $\tilde{\gamma}_p + \tilde{\gamma}_q$ is the horizontal lift of γ starting at $p + q$. Thus

$$\begin{aligned} m_{\lambda * p} \tilde{\gamma}'(0) &= (\lambda \tilde{\gamma}_p)'(0) \in \mathcal{H}_{\lambda p}, \\ a_{*(p,q)}(\tilde{\gamma}'_p(0), \tilde{\gamma}'_q(0)) &= (\tilde{\gamma}_p + \tilde{\gamma}_q)'(0) \in \mathcal{H}_{p+q}. \end{aligned} \quad \square$$

The above discussion is summarized in the following theorem.

Theorem 1.23. $\mathcal{H}$ is a horizontal distribution of E .

Horizontal Distribution determines Covariant Derivative

Conversaly, given a horizontal distribution $\mathcal{H}$, we can derive a connection ∇ on E as follows.

Definition 1.24. For any $p \in E$ and $u \in TE$, we can write $u = h(u) + v(u)$ where $h(u) \in \mathcal{H}$ and $v(u) \in \mathcal{V}$. We call $h(u)$ and $v(u)$ the horizontal and vertical component of u respectively.

For any $X \in C^\infty(M, TM)$ and $s \in C^\infty(M, E)$, the tangent map of s gives a map $s_* : TM \rightarrow TE$, then at any point $x \in M$ we can define $(\nabla_X s)(x)$ by

$$(\nabla_X s)(x) := \iota_{s(x)}^{-1} (v(s_{*x} X_x)).$$

Note that $\iota_{s(x)} : E_x \rightarrow \mathcal{V}_{s(x)}$ is the isomorphism defined above. The idea is taking the vertical component of $s_{*x}(X_x)$.

Theorem 1.25. ∇ defined above is a connection on E .

To prove this theorem, we need the following lemma.

Lemma 1.26. For $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u, w \in E_{\pi(p)}$, we have

$$\begin{aligned} m_{\lambda * p} \iota_p(u) &= \iota_{\lambda p}(\lambda u), \\ a_{*(p, q)}(\iota_p(u), \iota_q(w)) &= \iota_{p+q}(u + w). \end{aligned}$$

Proof. By the definition of ι_p ,

$$\begin{aligned} m_{\lambda * p} \iota_p(u) &= \left. \frac{d}{dt} \right|_{t=0} m_\lambda(p + tu) = \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\lambda u) = \iota_{\lambda p}(\lambda u), \\ a_{*(p, q)}(\iota_p(u), \iota_q(w)) &= \left. \frac{d}{dt} \right|_{t=0} (p + tu) + (q + tw) = \iota_{p+q}(u + w). \end{aligned} \quad \square$$

Lemma 1.27. For any $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ and $u \in TE_p$, $w \in TE_q$,

$$\begin{aligned} h(m_{\lambda * p} u) &= m_{\lambda * p}(h(u)), \quad v(m_{\lambda * p} u) = m_{\lambda * p}(v(u)), \\ h(a_{*(p, q)}(u, w)) &= a_{*(p, q)}(h(u), h(w)), \quad v(a_{*(p, q)}(u, w)) = a_{*(p, q)}(v(u), v(w)). \end{aligned}$$

Proof. Decompose u and $m_{\lambda * p} u$ into their horizontal and vertical components, we get

$$m_{\lambda * p}(h(u)) + m_{\lambda * p}(v(u)) = m_{\lambda * p}(u) = h(m_{\lambda * p} u) + v(m_{\lambda * p} u).$$

Since $\mathcal{H}$ is compatible with the linear structure of the vector bundle, we have $m_{\lambda * p}(h(u)) \in \mathcal{H}_{\lambda p}$. Besides, Lemma 1.26 implies $m_{\lambda * p}(v(u)) \in \mathcal{V}_{\lambda p}$, thus

$$m_{\lambda * p}(h(u)) = h(m_{\lambda * p} u), \quad m_{\lambda * p}(v(u)) = v(m_{\lambda * p} u).$$

The proof for the addition is exactly the same. $\square$

Lemma 1.28. For $\nu \in T_\lambda \mathbb{R} \cong \mathbb{R}$ and $u \in T_p E$, we have

$$m_{*(\lambda, p)}(\nu, u) = \nu \cdot \iota_p(p) + m_{\lambda * p} u.$$

Proof. let $\gamma(t)$ be a curve in E such that $\gamma(0) = p$ and $\gamma'(0) = u$, then

$$\begin{aligned} m_{*(\lambda, p)}(\nu, u) &= \left. \frac{d}{dt} \right|_{t=0} (\lambda + t\nu)\gamma(t) \\ &= \left. \frac{d}{dt} \right|_{t=0} (\lambda p + t\nu p) + \left. \frac{d}{dt} \right|_{t=0} \lambda \gamma(t) \\ &= \nu \cdot \iota_p(p) + m_{\lambda * p} u. \end{aligned} \quad \square$$

Proof of Theorem 1.25. We need to verify that ∇ satisfies the properties of a connection.

- $\nabla_{f_1 X_1 + f_2 X_2} s = f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s$:

$$\begin{aligned}
\nabla_{f_1 X_1 + f_2 X_2} s(x) &= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1 X_1 + f_2 X_2)) \right) \\
&= \iota_{s(x)}^{-1} \left(v(s_{*x}(f_1(x) X_{1x})) + v(s_{*x}(f_2(x) X_{2x})) \right) \\
&= \iota_{s(x)}^{-1} \left(f_1(x) v(s_{*x} X_{1x}) + f_2(x) v(s_{*x} X_{2x}) \right) \\
&= f_1 \nabla_{X_1} s(x) + f_2 \nabla_{X_2} s(x).
\end{aligned}$$

- $\nabla_X(\lambda_1 s_1 + \lambda_2 s_2) = \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2$:

For $\mathbb{R}$ -linearity,

$$\begin{aligned}
\nabla_X(\lambda s)(x) &= \iota_{(\lambda s)(x)}^{-1} \left(v((\lambda s)_{*x}(X_x)) \right) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(v(m_{\lambda * s(x)} s_{*x}(X_x)) \right) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(m_{\lambda * s(x)} v(s_{*x}(X_x)) \right) \quad (\text{By Lemma 1.27}) \\
&= \iota_{(\lambda s)(x)}^{-1} \left(\iota_{(\lambda s)(x)}(\lambda \nabla_X s(x)) \right) \quad (\text{By Lemma 1.26}) \\
&= \lambda \nabla_X s(x),
\end{aligned}$$

For additivity,

$$\begin{aligned}
\nabla_X(s_1 + s_2)(x) &= \iota_{(s_1 + s_2)(x)}^{-1} \left(v((s_1 + s_2)_{*x} X_x) \right) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(v(a_{*(s_1(x), s_2(x))}(s_{1*x} X_x, s_{2*x} X_x)) \right) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(a_{*(s_1(x), s_2(x))}(v(s_{1*x} X_x), v(s_{2*x} X_x)) \right) \quad (\text{By Lemma 1.27}) \\
&= \iota_{(s_1 + s_2)(x)}^{-1} \left(\iota_{(s_1 + s_2)(x)}(\nabla_X s_1(x) + \nabla_X s_2(x)) \right) \quad (\text{By Lemma 1.26}) \\
&= \nabla_X s_1(x) + \nabla_X s_2(x).
\end{aligned}$$

- $\nabla_X(fs) = df(X) \cdot s + f \nabla_X s$:

$$\begin{aligned}
\nabla_X(fs)(x) &= \iota_{(fs)(x)}^{-1} \left(v((fs)_{*x} X_x) \right) \\
&= \iota_{(fs)(x)}^{-1} \left(v(m_{f(x)*s(x)}(f_{*x} X_x, s_{*x} X_x)) \right) \\
&= \iota_{(fs)(x)}^{-1} \left(df(X)_x \cdot \iota_{(fs)(x)}(s(x)) + v(m_{f(x)*s(x)} s_{*x} X_x) \right) \quad (\text{By Lemma 1.28}) \\
&= \iota_{(fs)(x)}^{-1} \left(\iota_{(fs)(x)}(df(X)_x \cdot s(x) + f(x) \nabla_X s(x)) \right) \quad (\text{By Lemma 1.26}) \\
&= df(X)_x \cdot s(x) + f(x) \nabla_X s(x).
\end{aligned}$$

□

Exercises

Exercise 1.1. Given a connection ∇ , we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s) -tensor field T , the covariant derivative $\nabla_Z T$ is

$$\begin{aligned} (\nabla_Z T)(X_1, \dots, X_r, \omega^1, \dots, \omega^s) &:= Z(T(X_1, \dots, X_r, \omega^1, \dots, \omega^s)) \\ &\quad - \sum_{i=1}^r T(X_1, \dots, \nabla_Z X_i, \dots, X_r, \omega^1, \dots, \omega^s) \\ &\quad - \sum_{j=1}^s T(X_1, \dots, X_r, \omega^1, \dots, \nabla_Z \omega^j, \dots, \omega^s). \end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X, Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X, Y]} T.$$

We define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s+1)$ -tensor field

$$(\nabla_{\text{tensor}} T)(Z, \dots, X_i, \dots, \dots, \omega^j, \dots) := (\nabla_Z T)(\dots, X_i, \dots, \dots, \omega^j, \dots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

[Prove that](#) for a torsion-free connection, ∇_{tensor} satisfies **the Ricci identity**

$$(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) - (\nabla_{\text{tensor}}^2 T)(Y, X, \dots, \dots) = (R(X, Y)T)(\dots, \dots)$$

Exercise 1.2. For a vector field $Z \in C^\infty(M, TM)$, suppose ∇ is torsion-free. Both $\nabla^2 Z$ and $\nabla_{\text{tensor}}^2 Z$ are $(1, 2)$ -tensor fields, but by Ricci identity, we have

$$(\nabla_{\text{tensor}}^2 Z)(X, Y) - (\nabla_{\text{tensor}}^2 Z)(Y, X) = R(X, Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?

Exercise 1.3 (Torsion 2-form). Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. Define

$$\tau^i := d\theta^i + \omega_j^i \wedge \theta^j.$$

If we arrange $\theta = (\theta^1, \dots, \theta^n)^T$ as a column vector, then the above formula can be written as

$$\tau = d\theta + \omega \wedge \theta.$$

then τ is called the **torsion 2-form** of ∇ w.r.t. $\{e_i\}$.

Let $T(e_i, e_j) = T_{ij}^k e_k$, where T is the torsion of ∇ . [Prove that](#)

$$\tau^k(e_i, e_j) = T_{ij}^k.$$

Thus

$$\tau^k = \frac{1}{2} T_{ij}^k \theta^i \wedge \theta^j.$$

Exercise 1.4 (Levi-Civita connection). Let (M, g) be a Riemannian manifold, ∇ be a connection on TM . Show that

1. ∇ is torsion-free if and only if $\tau = 0$, i.e.,

$$d\theta^i + \omega_j^i \wedge \theta^j = 0.$$

2. If $\{e_i\}$ is a local orthonormal frame, then ∇ is compatible with g if and only if ω is skew-symmetric, i.e.,

$$\omega_j^i + \omega_i^j = 0.$$

3. Under a local orthonormal frame, there exists a unique connection 1-form ω satisfying

$$d\theta^i + \omega_j^i \wedge \theta^j = 0, \quad \omega_j^i + \omega_i^j = 0.$$

This is the existence and uniqueness of the Levi-Civita connection.

Exercise 1.5. Let ∇ be a connection on TM , $\{e_i\}$ be a local frame of TM , $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. The following two equations

$$d\theta = \tau - \omega \wedge \theta,$$

$$d\omega = \Omega - \omega \wedge \omega,$$

are called the **structure equations**. Show that τ and Ω satisfy

$$d\tau = \Omega \wedge \theta - \omega \wedge \tau,$$

$$d\Omega = \Omega \wedge \omega - \omega \wedge \Omega.$$

These two equations are called the **Bianchi identities**. Furthermore, show that

$$d(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

Exercise 1.6. Show that if ∇ is torsion-free, then $d\tau = \Omega \wedge \theta - \omega \wedge \tau$ is equivalent to the **first Bianchi identity**

$$R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0.$$

Exercise 1.7. Show that if ∇ is torsion-free, then $d\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is equivalent to the **second Bianchi identity**

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Exercise 1.8. For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega_j^k = \Gamma_{ij}^k dx^i$, show that the transformation formula

$$\tilde{\omega} = a^{-1} da + a^{-1} \omega a$$

reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial \tilde{x}^k}{\partial x^l} \frac{\partial x^m}{\partial \tilde{x}^i} \frac{\partial x^n}{\partial \tilde{x}^j} \Gamma_{mn}^l + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j} \frac{\partial \tilde{x}^k}{\partial x^l}.$$

Exercise 1.9. Let ∇ be a connection on TM . It determines a horizontal distribution $\mathcal{H}$ on TM . Then $T(TM) = \mathcal{V} \oplus \mathcal{H}$, where $\mathcal{V}$ is the vertical bundle of TM . Show that ∇ is torsion-free if and only if for any $X, Y \in C^\infty(M, TM)$,

$$[X^h, Y^h] = [X, Y]^h.$$